In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
```

In [2]:
```
!pip install statsmodels
```

```
Collecting statsmodels
  Downloading statsmodels-0.15.0-cp314-cp314-macosx_11_0_arm64.whl.metadata
(9.6 kB)
Requirement already satisfied: numpy<3,>=1.23.5 in /opt/miniconda3/lib/pytho
n3.14/site-packages (from statsmodels) (2.5.2)
Requirement already satisfied: scipy!=1.9.2,>=1.8 in /opt/miniconda3/lib/pyt
hon3.14/site-packages (from statsmodels) (1.18.1)
Requirement already satisfied: pandas!=2.1.0,>=1.4 in /opt/miniconda3/lib/py
thon3.14/site-packages (from statsmodels) (3.0.5)
Collecting patsy>=0.5.6 (from statsmodels)
  Downloading patsy-1.0.3-py2.py3-none-any.whl.metadata (3.7 kB)
Requirement already satisfied: packaging>=21.3 in /opt/miniconda3/lib/python
3.14/site-packages (from statsmodels) (26.0)
Collecting formulaic>=1.1.0 (from statsmodels)
  Downloading formulaic-1.2.2-py3-none-any.whl.metadata (7.0 kB)
Collecting interface-meta>=1.2.0 (from formulaic>=1.1.0->statsmodels)
  Downloading interface_meta-2.0.1-py3-none-any.whl.metadata (6.4 kB)
Requirement already satisfied: narwhals>=1.17 in /opt/miniconda3/lib/python
3.14/site-packages (from formulaic>=1.1.0->statsmodels) (2.25.0)
Requirement already satisfied: typing-extensions>=4.2.0 in /opt/miniconda3/l
ib/python3.14/site-packages (from formulaic>=1.1.0->statsmodels) (4.16.0)
Collecting wrapt>=1.17.0rc1 (from formulaic>=1.1.0->statsmodels)
  Downloading wrapt-2.5.0-cp314-cp314-macosx_11_0_arm64.whl.metadata (7.6 k
B)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/miniconda3/li
b/python3.14/site-packages (from pandas!=2.1.0,>=1.4->statsmodels) (2.9.0.po
st0)
Requirement already satisfied: six>=1.5 in /opt/miniconda3/lib/python3.14/si
te-packages (from python-dateutil>=2.8.2->pandas!=2.1.0,>=1.4->statsmodels)
(1.17.0)
Downloading statsmodels-0.15.0-cp314-cp314-macosx_11_0_arm64.whl (11.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 11.6/11.6 MB 7.0 MB/s  0:00:016.
8 MB/s eta 0:00:0101
Downloading formulaic-1.2.2-py3-none-any.whl (118 kB)
Downloading interface_meta-2.0.1-py3-none-any.whl (15 kB)
Downloading patsy-1.0.3-py2.py3-none-any.whl (233 kB)
Downloading wrapt-2.5.0-cp314-cp314-macosx_11_0_arm64.whl (106 kB)
Installing collected packages: wrapt, patsy, interface-meta, formulaic, stat
smodels
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5/5 [statsmodels][0m 4/5 [statsm
odels]
Successfully installed formulaic-1.2.2 interface-meta-2.0.1 patsy-1.0.3 stat
smodels-0.15.0 wrapt-2.5.0
```

In [2]:
```
data = pd.read_csv(
    'daily-website-visitors.csv',
    parse_dates=['Date'],
```

```
    index_col='Date'
)
```

In [3]:
```
data['Unique.Visits'] = pd.to_numeric(
    data['Unique.Visits'], errors='coerce'
)

data = data.dropna(subset=['Unique.Visits'])
data = data.sort_index()

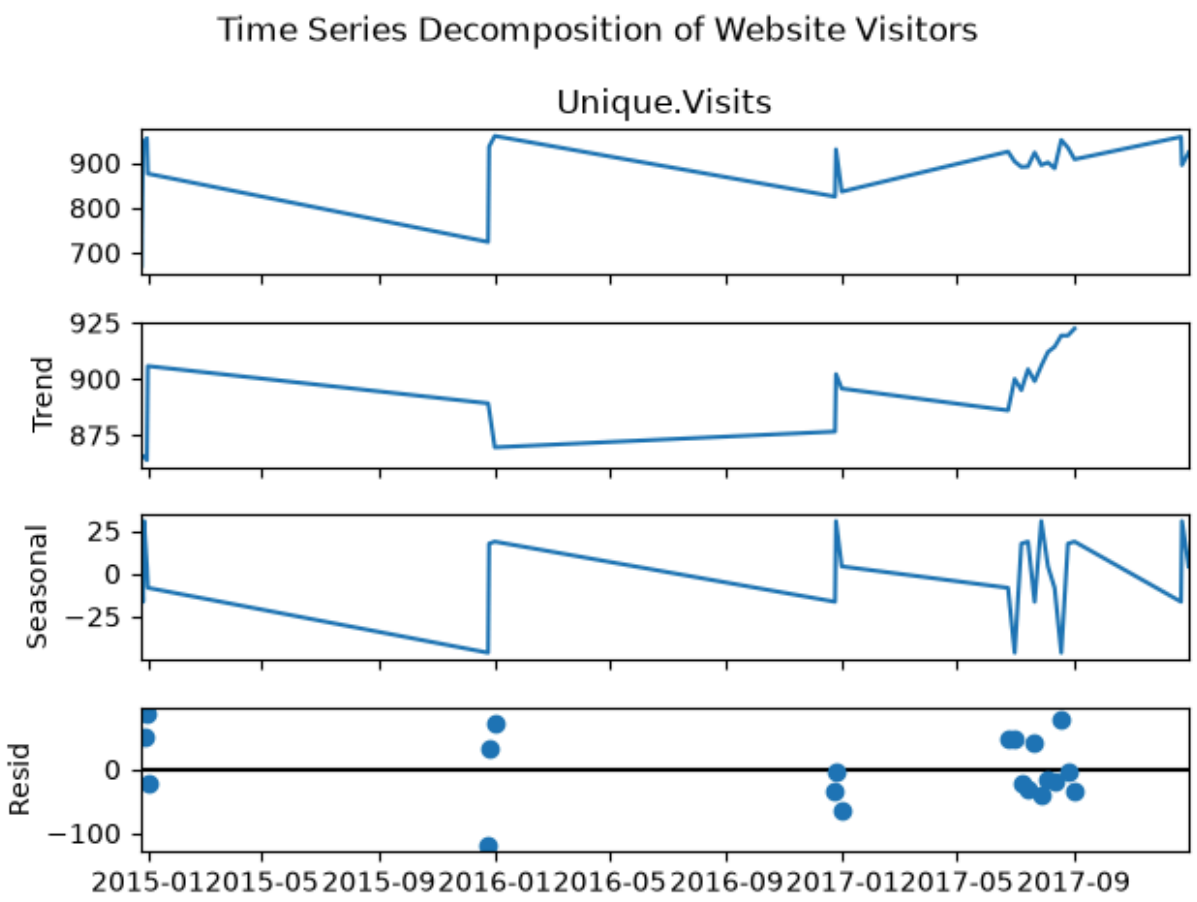
print(data.head())
print(data.info())
```

```
             Row        Day  Day.Of.Week Page.Loads  Unique.Visits  \
Date
2014-12-24  102  Wednesday            4      1,430          947.0
2014-12-25  103   Thursday            5      1,002          667.0
2014-12-27  105   Saturday            7      1,345          941.0
2014-12-28  106     Sunday            1      1,453          948.0
2014-12-31  109  Wednesday            4      1,381          955.0

            First.Time.Visits Returning.Visits
Date
2014-12-24                772              175
2014-12-25                522              145
2014-12-27                781              160
2014-12-28                765              183
2014-12-31                773              182
<class 'pandas.DataFrame'>
DatetimeIndex: 26 entries, 2014-12-24 to 2017-12-31
Data columns (total 7 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Row                26 non-null     int64
 1   Day                26 non-null     str
 2   Day.Of.Week        26 non-null     int64
 3   Page.Loads         26 non-null     str
 4   Unique.Visits      26 non-null     float64
 5   First.Time.Visits  26 non-null     str
 6   Returning.Visits   26 non-null     str
dtypes: float64(1), int64(2), str(4)
memory usage: 1.6 KB
None
```

In [4]:
```
decomposition = seasonal_decompose(
    data['Unique.Visits'],
    model='additive',
    period=7
)

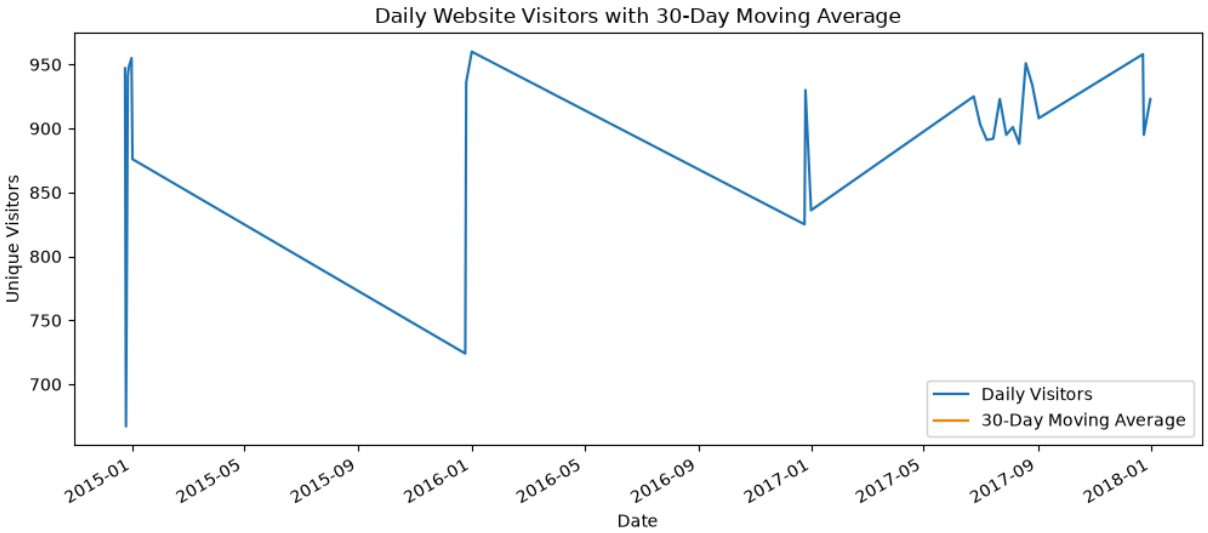
decomposition.plot()
plt.suptitle('Time Series Decomposition of Website Visitors')
plt.tight_layout()
plt.show()
```



In [5]:
```
data['Visitors_MA'] = data['Unique.Visits'].rolling(window=30).mean()

data[['Unique.Visits', 'Visitors_MA']].plot(figsize=(12, 5))

plt.title('Daily Website Visitors with 30-Day Moving Average')
plt.xlabel('Date')
plt.ylabel('Unique Visitors')
plt.legend(['Daily Visitors', '30-Day Moving Average'])
plt.show()
```



In [6]:
```
data['Visitors_Z'] = (
    (data['Unique.Visits'] - data['Unique.Visits'].mean())
    / data['Unique.Visits'].std()
)

anomalies = data[data['Visitors_Z'].abs() > 3]

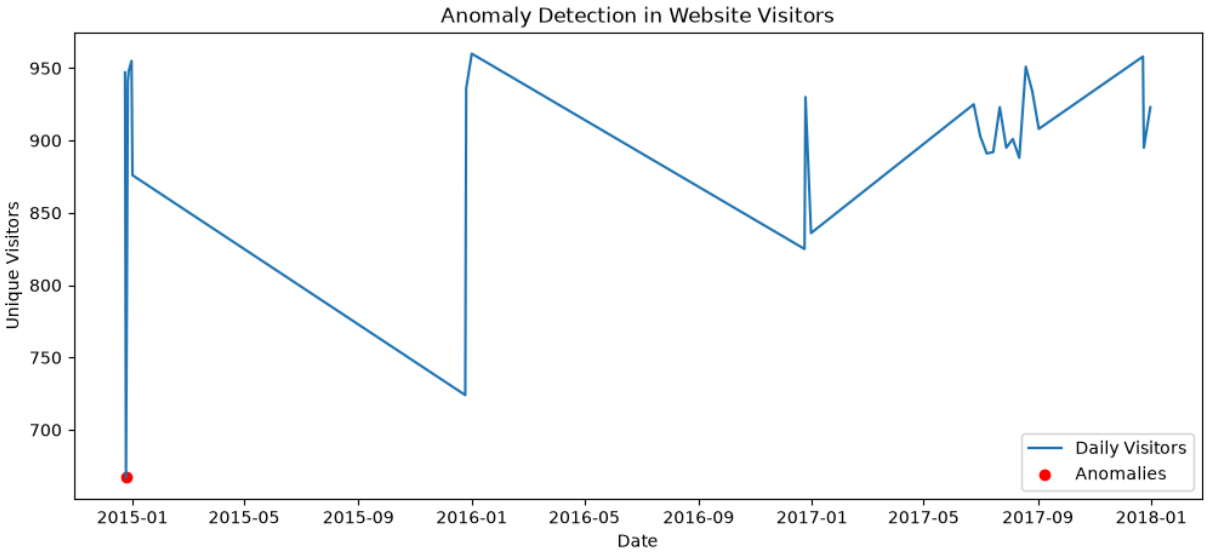
plt.figure(figsize=(12, 5))

plt.plot(
    data.index,
    data['Unique.Visits'],
    label='Daily Visitors'
)

plt.scatter(
    anomalies.index,
    anomalies['Unique.Visits'],
    color='red',
    label='Anomalies'
)

plt.title('Anomaly Detection in Website Visitors')
plt.xlabel('Date')
plt.ylabel('Unique Visitors')
plt.legend()
plt.show()

print("Detected anomalies:")
print(anomalies[['Unique.Visits', 'Visitors_Z']])
```




```
Detected anomalies:
            Unique.Visits  Visitors_Z
Date
2014-12-25          667.0   -3.330647
```

In [ ]: